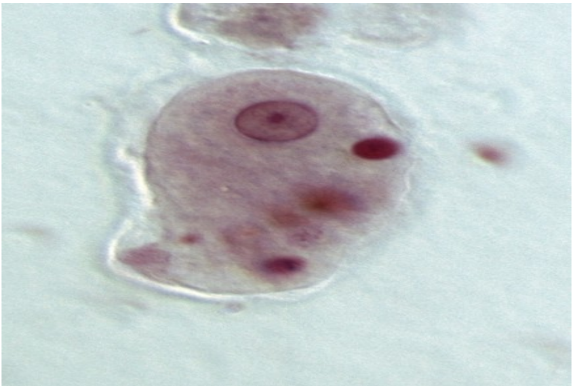
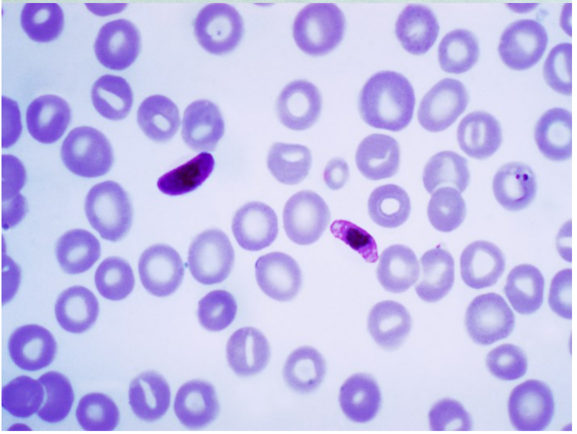
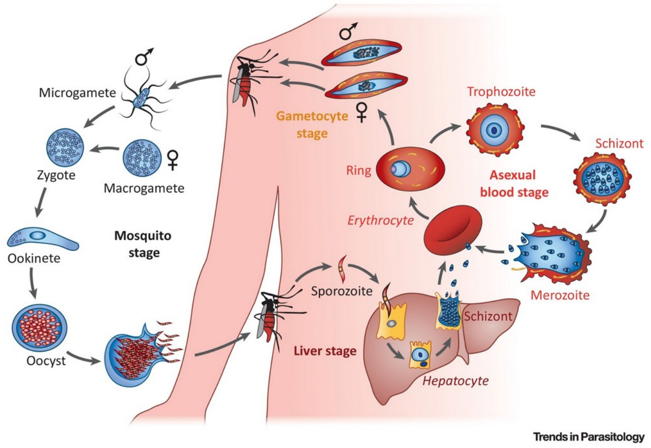




Protozoa

Lecturer:: Asmaa Khalis Mohammed

Microbiology department



General Properties of Protozoa

- Protozoa, proto = first, zoa = animal. Single celled organisms contain organelles to carry specific function.
- Protozoa include many distinct lineages of chemoheterotrophic unicellular or colonial organisms. A few species are pathogenic in nature parasitize human and other animals, causing hundreds of millions of infections in a year around the world.

- They have a relatively complex internal structure and carry out complex metabolic activities such as digestion, reproduction, respiration, excretion, etc.
- Mostly unicellular organism with fully functional cell, No **organs** or **tissues** are formed but specialized organelles serve many of these functions.
- Include Live freely, may be **parasitic, symbiotic, commensal and mutualistic** forms.

- They are motile have locomotive organelles. e.g flagella ***Trypanosoma, Leishmania***), and cilia for movement, pseudopodia (***Entamoeba coli***) and cilia (***Balantidium coli***).
- Reproduction is asexual by binary fission (***Entamoeba histolytica, E.coli & Balantidium coli***, cysts (***E.histolytica***) or sexual by conjugation (***B. coli***).
- All types of nutrition are present chemohetrotrophs : **autotrophic** or **phototrophic** (plants like), **heterotrophic** (animal like).

Morphology

- Protozoa are Eukaryotic resemble to animal cell, contain major cell organelles (including nucleus and mitochondria).
- There organelles are highly specialized for **feeding**, **reproduction** and **movement**.
- It is the vital structure of a cell & present in the cytoplasm.
- Nucleus is surrounded by nuclear membrane, contains a **karyosome** and chromatin granules.
- **Karyosome** a small mass of chromatin, also called **endosome** or "**centrosome**."
- The cytoplasm of protozoa are divided into an outer layer called ***Ectoplasm*** and an inner layer called ***Endoplasm***.
- **Ectoplasm** helps in feeding, movement and protection.

- **Endoplasm** houses nucleus, mitochondria and food.
- Some protozoa have special **appendages flagella** or **cilia** that helps their movement.
- Freshwater protozoa have **contractile vacuoles** to pump out excess water.
- Their shape remain constant (specially **ciliates**) or change constantly (**Amoeba**).
- Except **sporozoites**, all type of protozoa are motile either through **flagella**, **cilia** or **pseudopodia**.
- Have **Eyespot** that can detect change in light. Respond to light and **learn**, by **trial & error**.

Introduction

- Medical parasitology is a study of invertebrate animals and diseases they cause.
- The most important protozoan genera:

Hemoflagellates	Luminal (GIT,GUT)	Blood/tissue
Leishmania sp,	***Giardia sp,***	***Plasmodium sp,***
Trypanosoma sp,	**Entamoeba sp,**	***Toxoplasma sp,***
	Cryptosporidium sp,	Babesia sp,
	Isospora sp,	
	Trichomonas sp,	

Classification of protozoa

- Protozoa classified on the basis of their **motility** & the method of reproduction.
- ***They are classified in to four main types***:
  - **Flagellates**
  - **Ciliates**
  - **Sarcodina**
  - **Sporozoates**

- **Pseudopodia:** A major loco-motor organelles in Sarcodina (***Entamoeba*).**
- **Flagellates:** They moved by help of flagella (a tail-like structure). The movement is thin protein whip, rotates or whips back and forth .(***Trypanosoma, Leishmania, Giardia, Trichomonas*).**
- **Ciliates:** Hair like protein fibers, Movement and attachment (***Balantidium coli***).
- **Sporozoates** are the only **non-motile** form of protozoa & **well developed** sexual and asexual stages.***Plasmodium spp, Toxoplasma gondii*** (causes **Toxoplasmosis**).

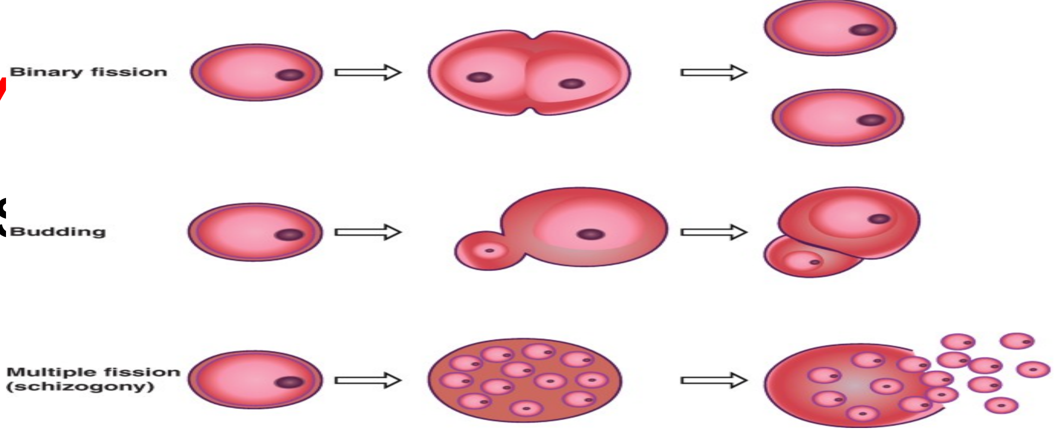
Reproduction in protozoa

Protozoa can reproduce their off spring by both sexual & asexual methods:

Asexual reproduction:

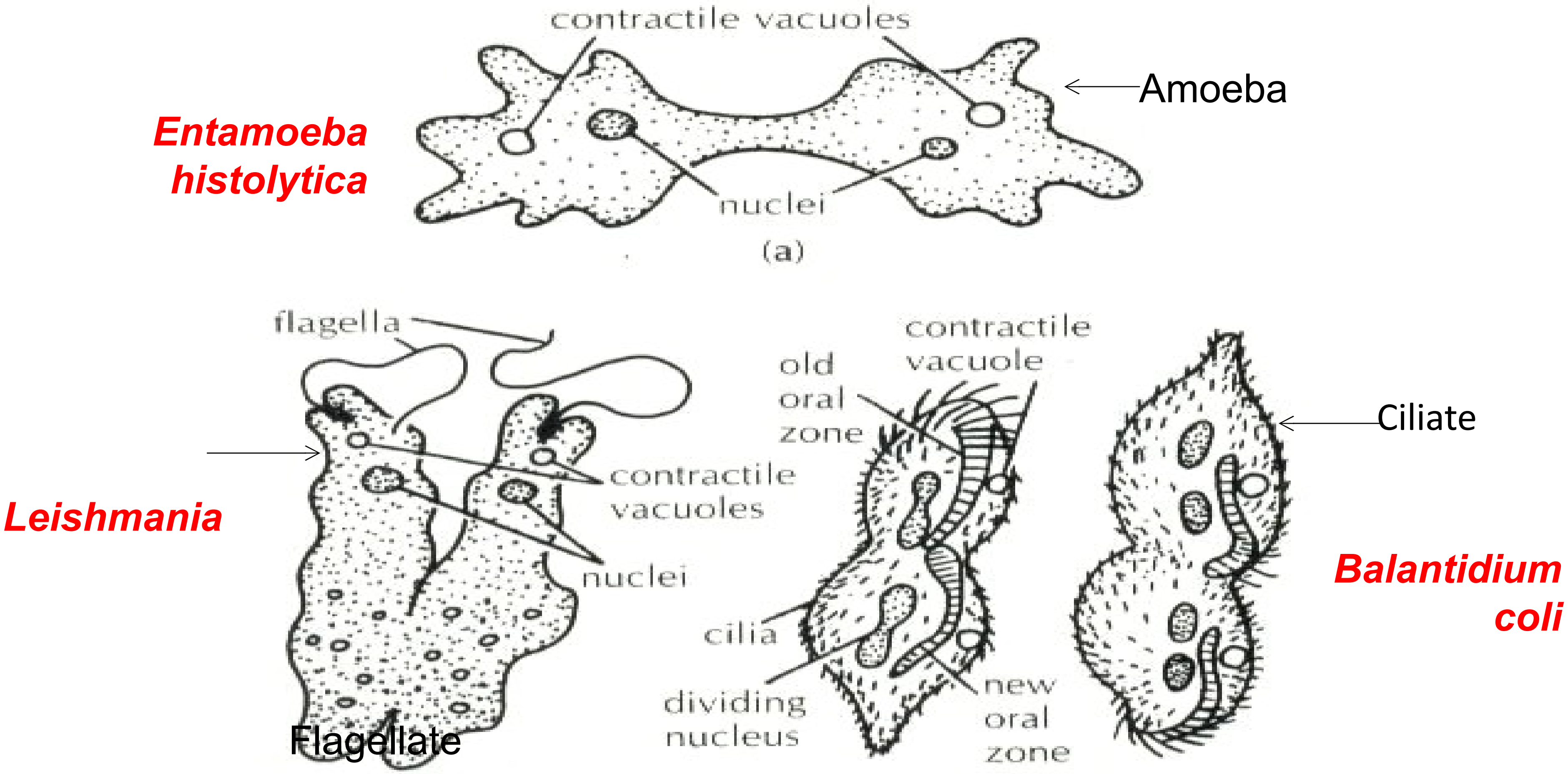
Asexual methods:

- **Simple binary fission:** The parasite reproduces by a simple division either longitudinally(***Try*** transversely into two equal parts



- **Budding**
- **Schizogony or multiple fission**

Asexual Reproduction



❖ Schizogony

- It is the method of **multiple fission** in which the first nucleus under goes multiple division, form many nuclei that a small portion of cytoplasm concentrate around each nucleus and then protozoan cell is divided in to many daughter cells.

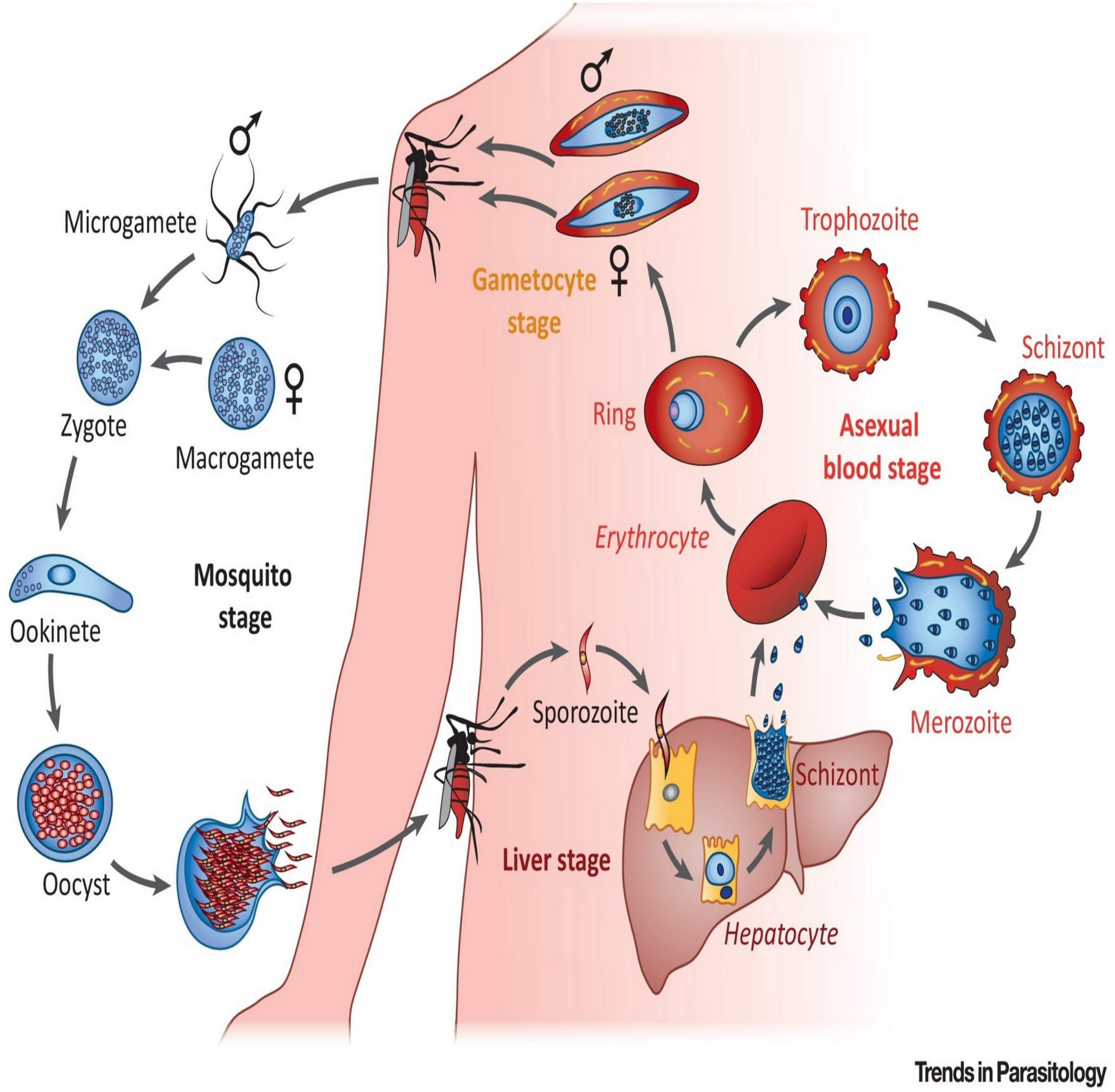
Sexual reproduction

❖Conjugation:

- Two protozoa meet together and exchange their genetic material.

❖Gametogony:

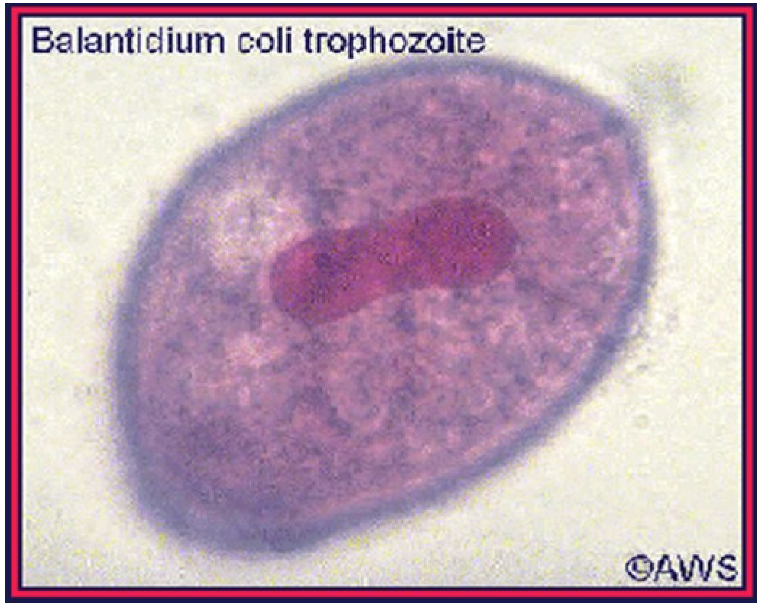
- Union of two sexually differentiated cells (zygote).

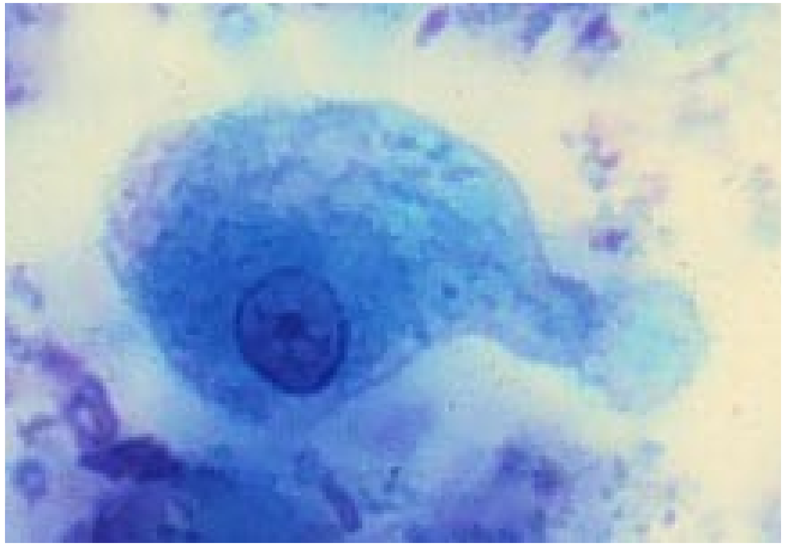


:Stages of protozoa (LIFE STAGES)

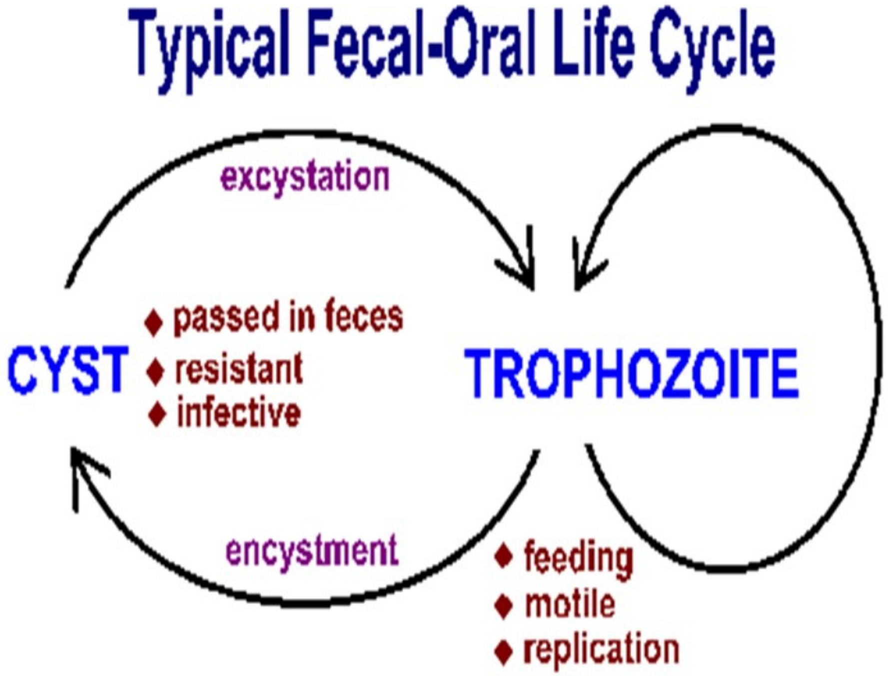
A protozoan parasite during its **life cycle** may exist in **two stages**: **trophozoite** and **cyst**.

- **Trophozoite:** The motile vegetative, active feeding stage, multiplies via binary fission; (e.g. ***Entamoeba, Balantidium***).



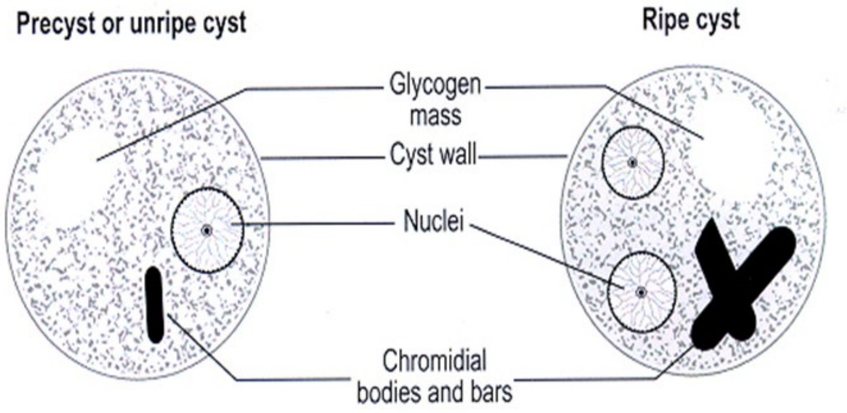


Entamoeba histolytica
Trophozoite

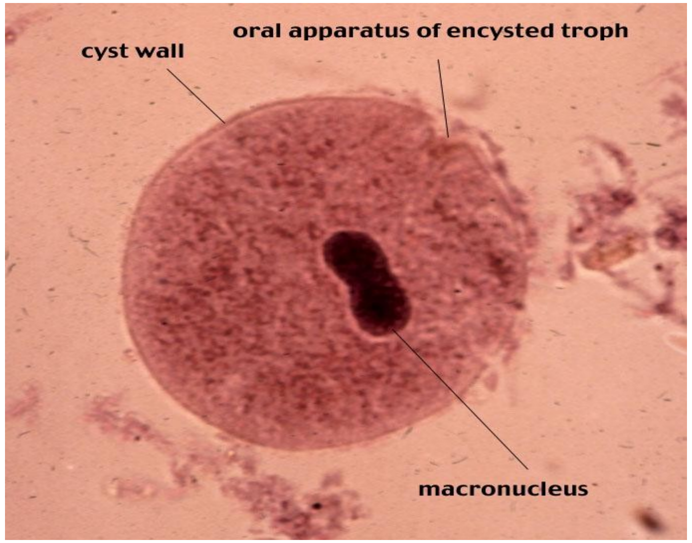


❖ Cyst:

Cyst is resistant form of the protozoa, inactive, non-motile, **infective stage**, survives the environment due to the presence of a cyst wall. some organisms divide within the .cyst wall. (e.g. ***Entamoeba histolytica***)



Entamoeba histolytica



Balantidium coli, cyst
Infective stage

Life cycle:

Life cycle of protozoa may be simple or complex.

1- Simple life cycle:

The life cycles are simple and completed in a single host only. The trophozoite replicate asexually by binary fission, e.g. ***Entamoeba, Giardia***.

2- Complex life cycle:

Both sexual & asexual reproduction take place in these parasites(**apicomplex**). These parasites require two different hosts. In ***Plasmodium*** species, sexual replication occurs in one host (***Anopheles,*** **mosquito**) and asexual replication in .another host (**human**)

Hosts

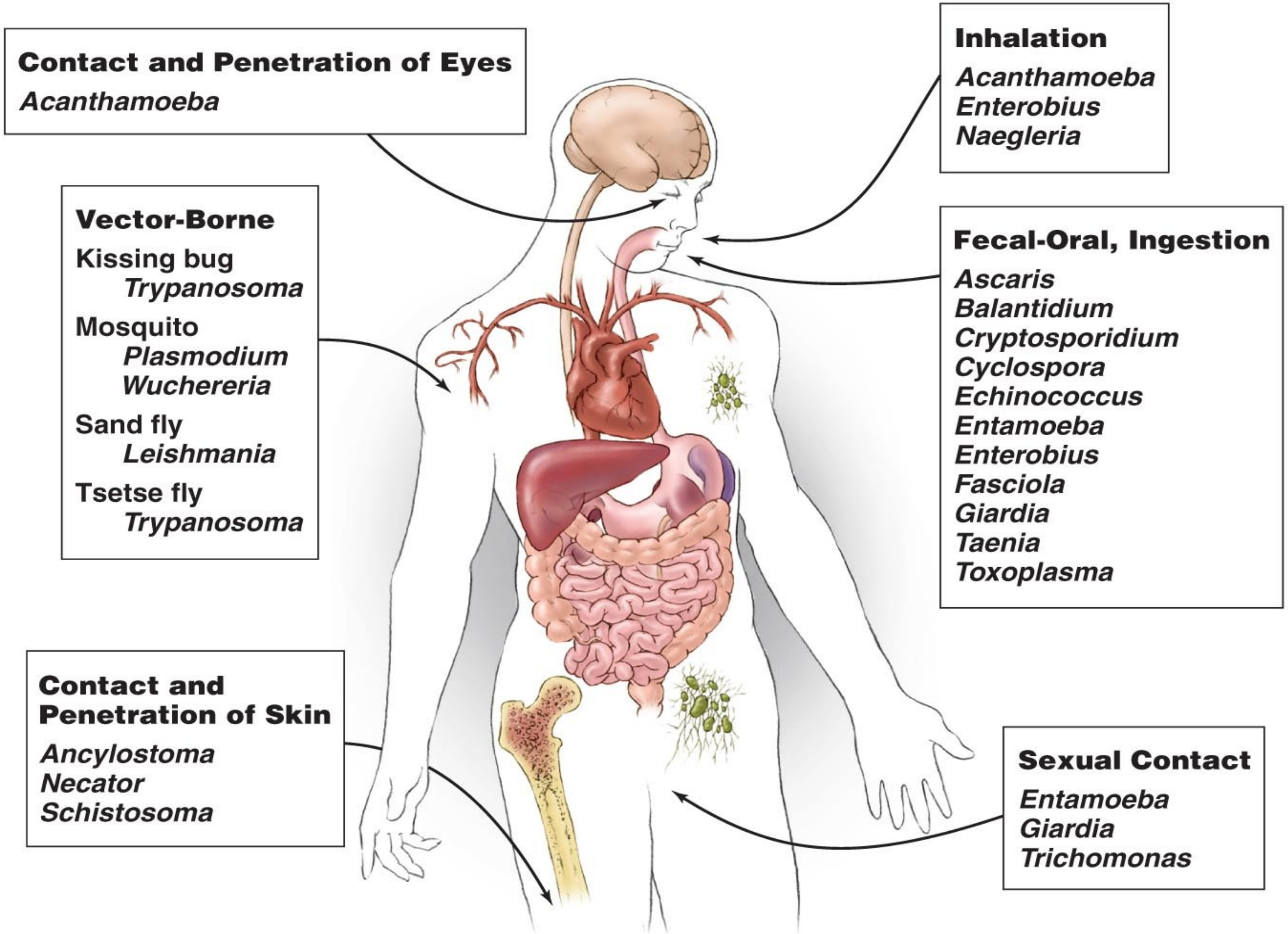
Infected host classified as:

- **Intermediate-**host in which **asexual stages** developed.
- **Definitive-**host in which **sexual stages** occur.

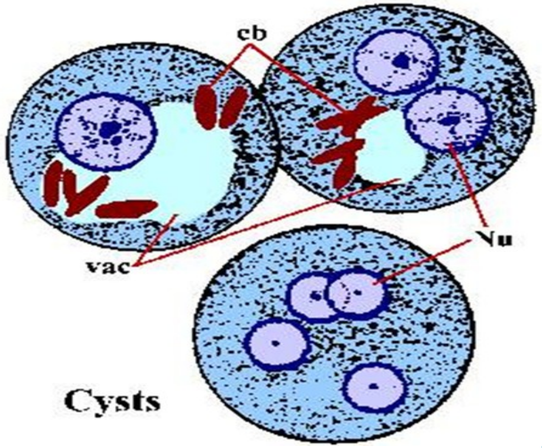
Vectors

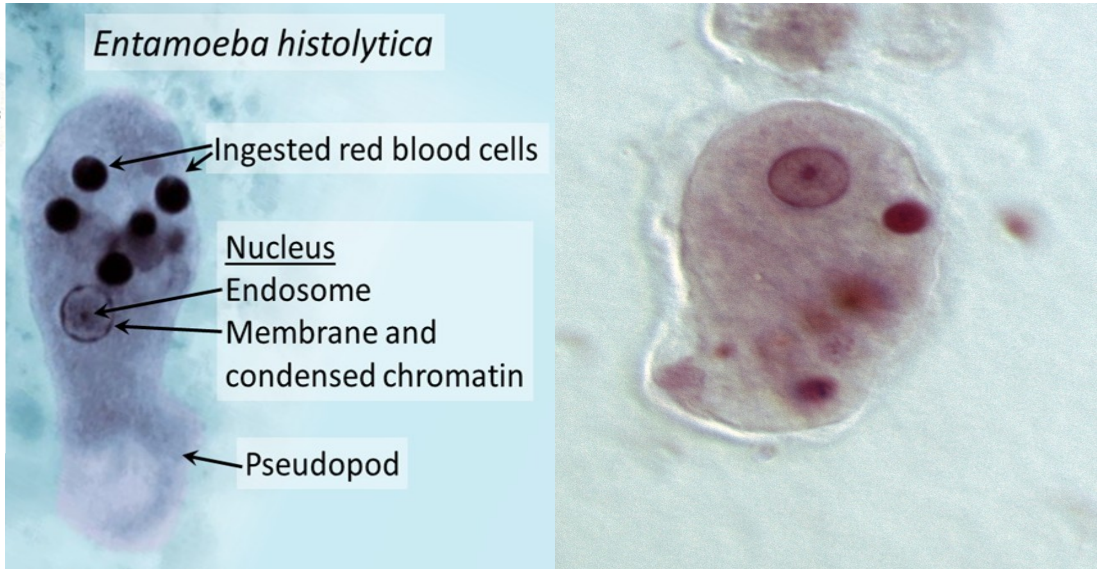
- Are living transmitters (a fly) of disease, may be;
- **Mechanical:** which trans port the parasite but there is nondevelopment of the parasite in the vector.
- **Biologic:** in which some stages of the life cycle occur.

Routes by which humans acquire parasitic infections



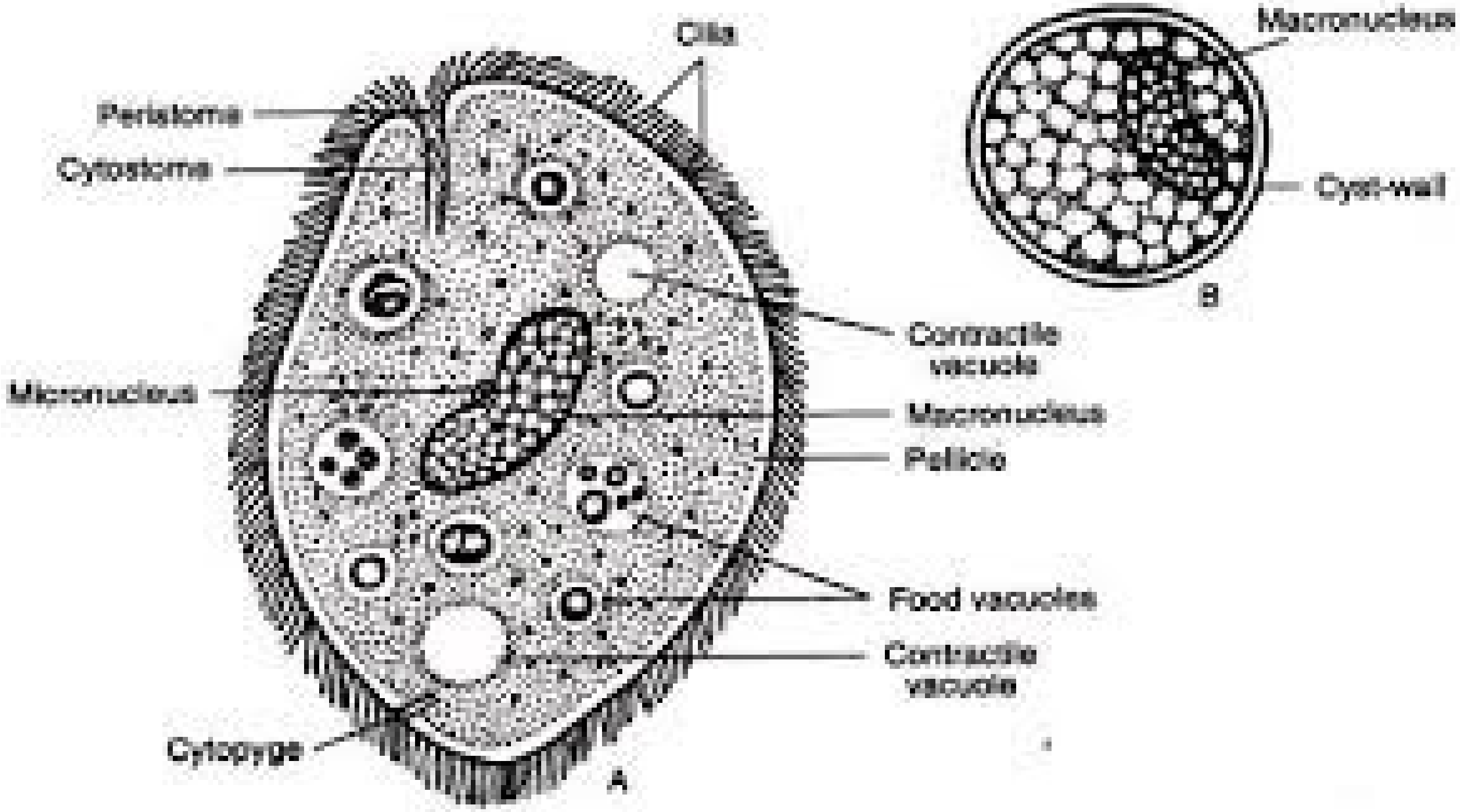
Entamoeba histolytica



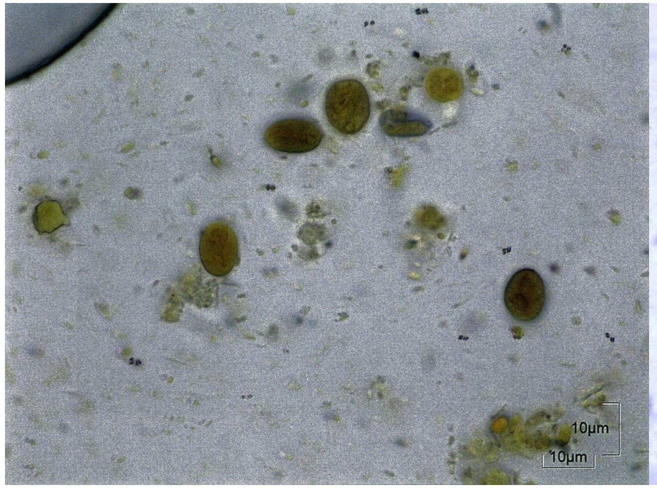


- **Life stages- Cyst & trophozoite.**
- **Transmission-** faecal-oral transmission, fresh fruits, and vegetables.
- **Diagnosis-** trophozoite or cysts in stool or serology.
- **Amebiasis-** dysentery; **inverted flask**-shaped lesions in large intestine which extension to peritoneum and liver, lungs, brain and heart. Blood and pus in stools and abscesses.

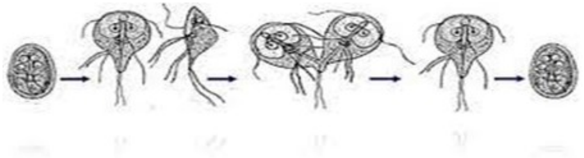
- ***Balantidium coli:*** Only a single nucleus is present in most of the protozoa but the ciliates have two nuclei, a small nucleus (**micronucleus**) for cell division and **macronucleus**) regulates metabolic activities.



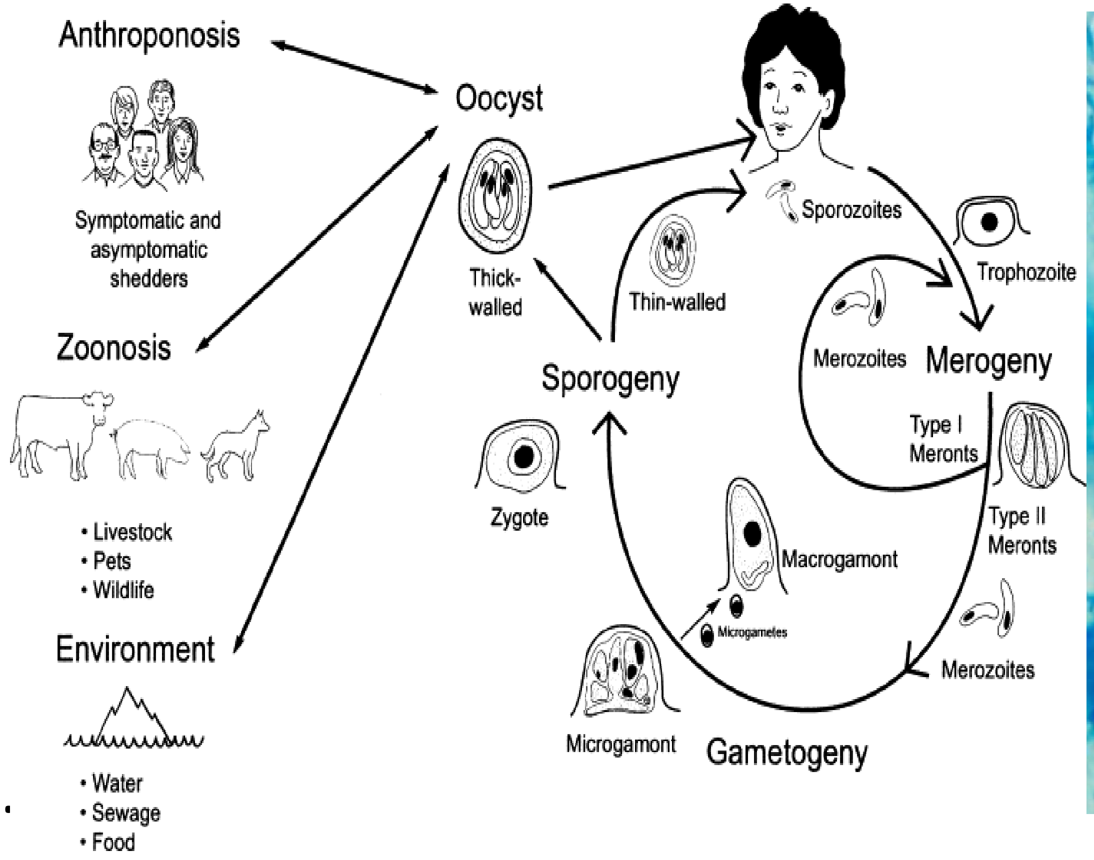
Giardia lambelia

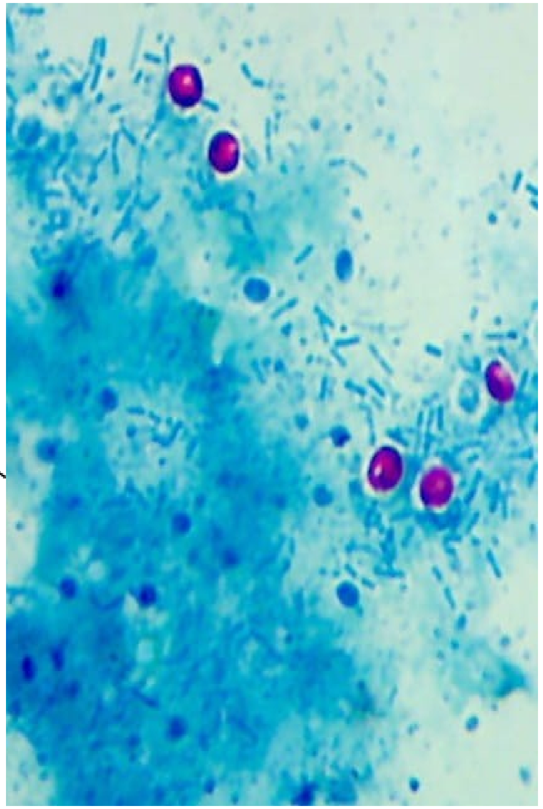


- **Life stages- cyst & trophozoite.**
- **Transmission-** Faecal (human, beaver, muskrat, etc). Oral transmission- water, water, food, day care, oral-anal sex.
- **Diagnosis-** trophozoites and cyst in stool or faecal antigen test (replaces string test).
- **Giardiasis**- ventral sucking disk attached to the lining of duodenal wall, causing **a fatty, foul smelling diarrhoea** (diarrhoea which leads to **malabsorption** duodenum, jejunum).

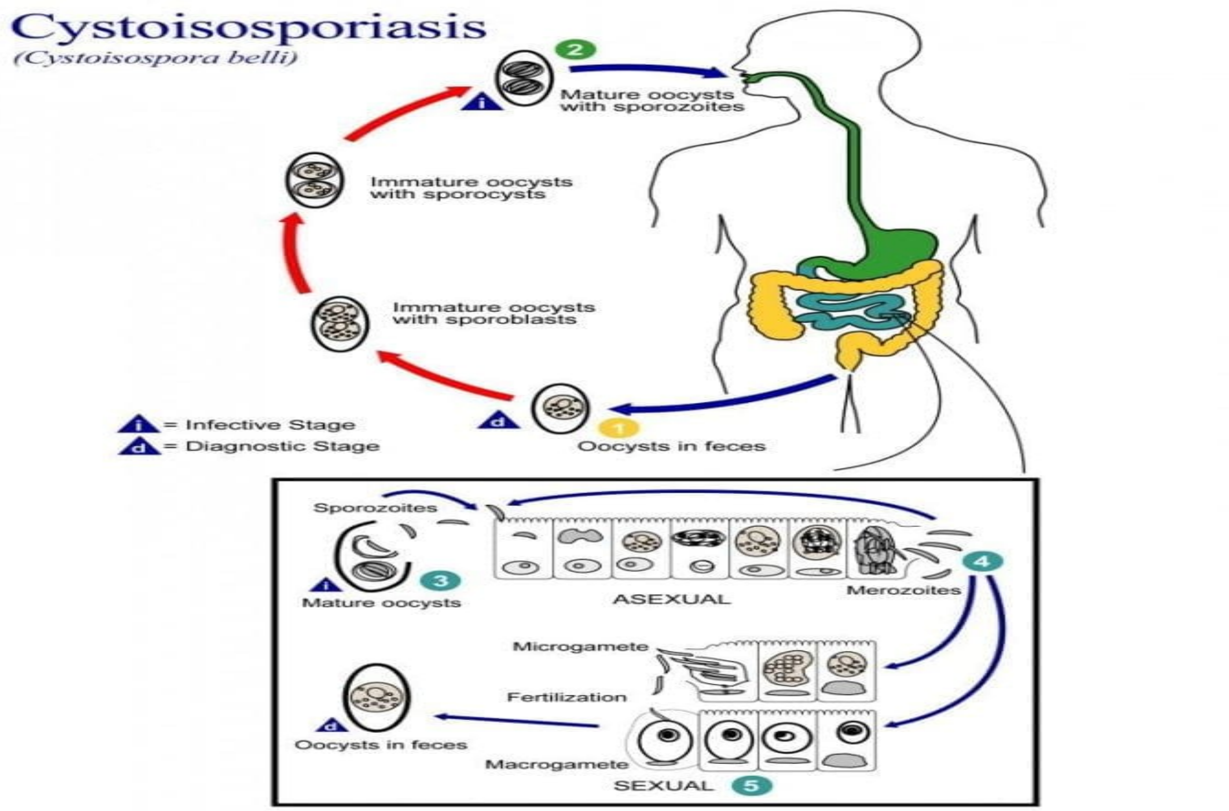


Cryptosporidium sp.





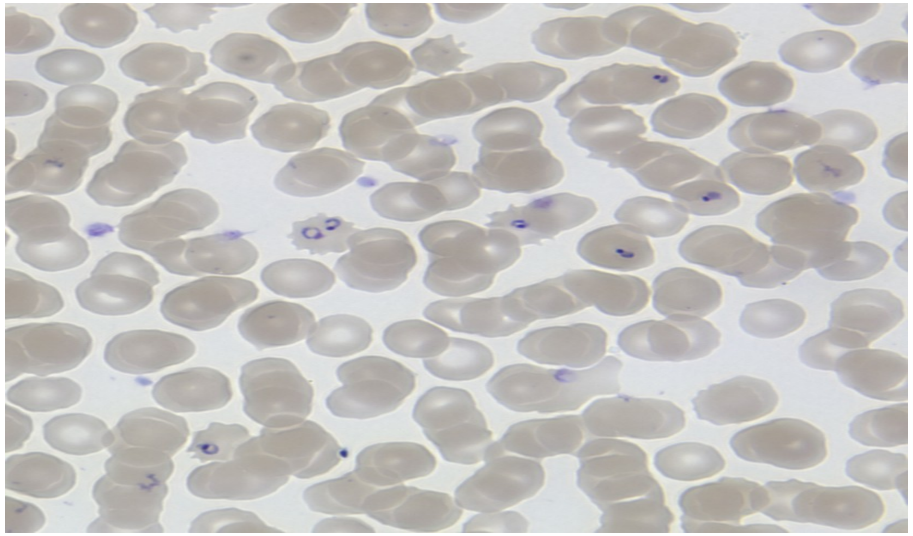
- ***Cryptosporidium parvum***
- **Life stages**-cyst & trophozoite.
- **Transmission**- undercooked meat, water; not killed by chlorination.
- **Diagnosis**- Acid fast staining- biopsy shows dots (cysts) in intestinal glands.
- **Cryptosporidiosis**-transient diarrhoea in healthy, sever in immunocompromised hosts.



- ***Isospora belli***
- **Life stage-** oocyst
- **Transmission**- ingestion faecal-oral, water.
- **Diagnosis-** Acid-fast and elliptical oocysts in stool; contain 2 sporocyst with each 4 sporozoites.
- **Transient diarrhoea** in ADIS patients; diarrhoea mimics giardiasis malabsorption syndrome.

Plasmodium Spp,

- Plasmodium has two distinct Hosts:
  - A vertebrate: such as human where asexual phase (**schizogony**) takes place in the liver and red blood cells.
  - An arthropod host (***Anopheles* mosquito**) where **gametogony** (sexual phase) and sporogony tale place.



Human red blood cells, some infected with the species Plasmodium falciparum